

C-DOT · CONTROL SCIENCE

Causal Forecasting Phase 1/2 Evidence

Five model families · 96 traffic groups · fail-closed selection

AUTHORITATIVE DECISION

No challenger promoted.

Protected seed 46003 remains untouched.

Executive scorecard

PHASE 1

**17/17 interface hashes match
causal metadata + packaging + seed locks**

BEST ACCURACY

**LightGBM · 12.51% WAPE
11.64% better than moving average**

EVENT BENEFIT

**27.54% lower peak underprediction
all candidate p90 coverage calibrated**

PROMOTION DECISION

**0/5 candidates eligible
seed 46003 not consumed**

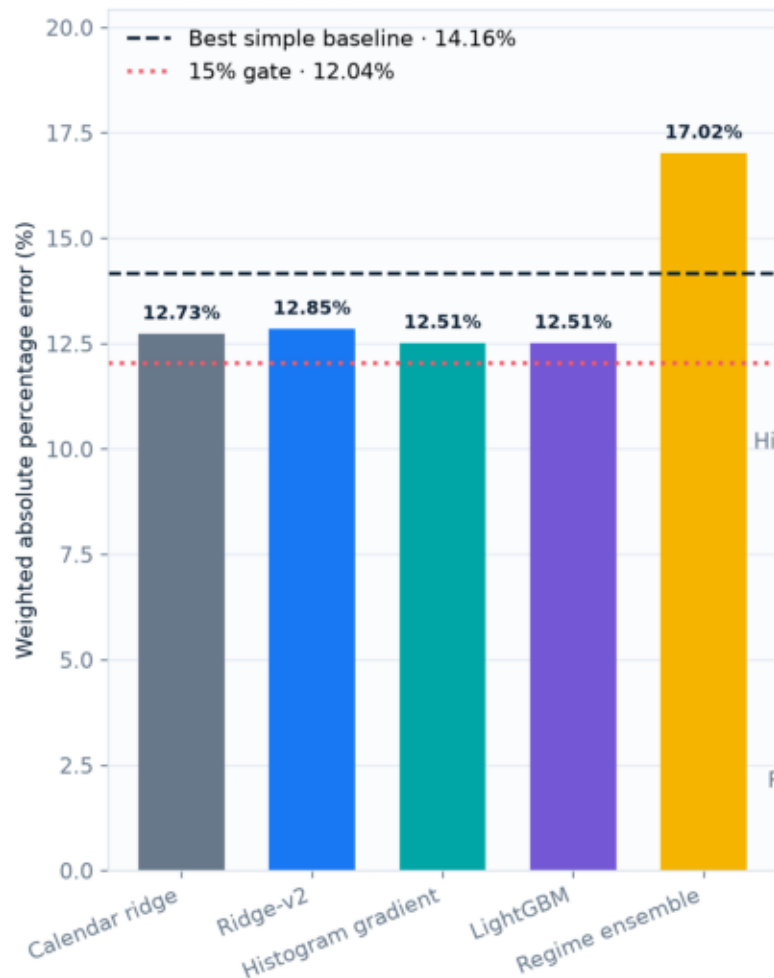
WHY NO PROMOTION

Best WAPE gain was below 15%; every model exceeded the 5% worst-slice regression guardrail.

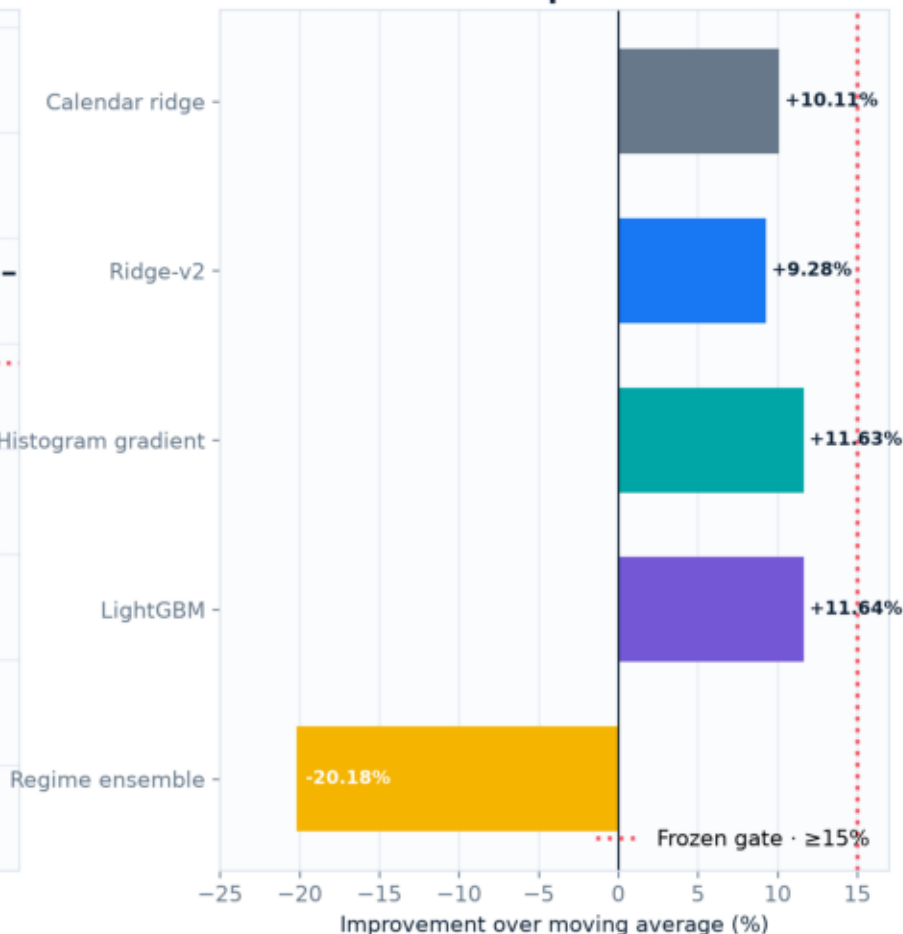
Forecast accuracy improved—but not enough to earn the protected test

Five frozen families · 96 groups · seed 46002 second-half selection · lower WAPE is better

Held-out WAPE

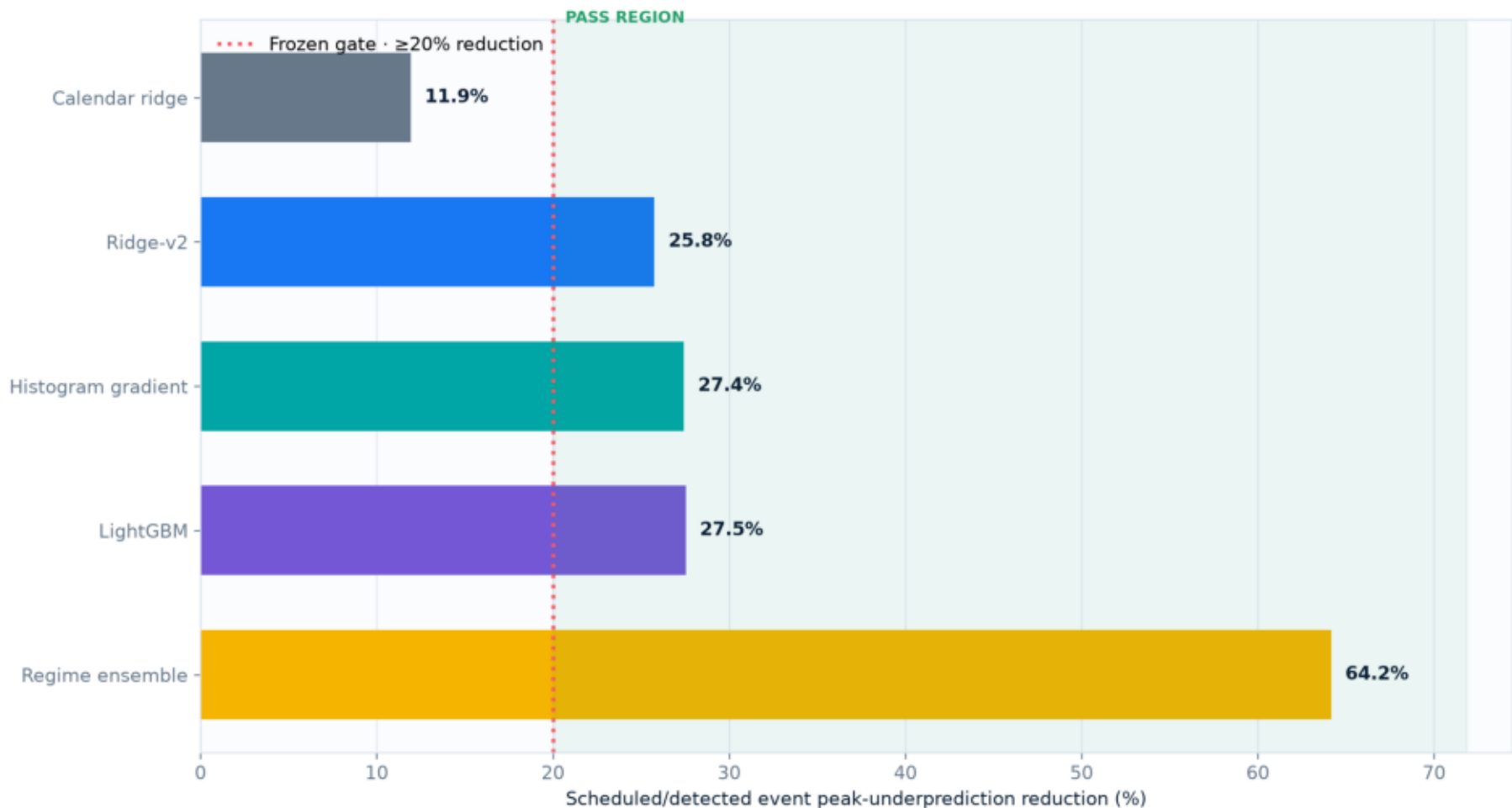


Relative to best simple baseline



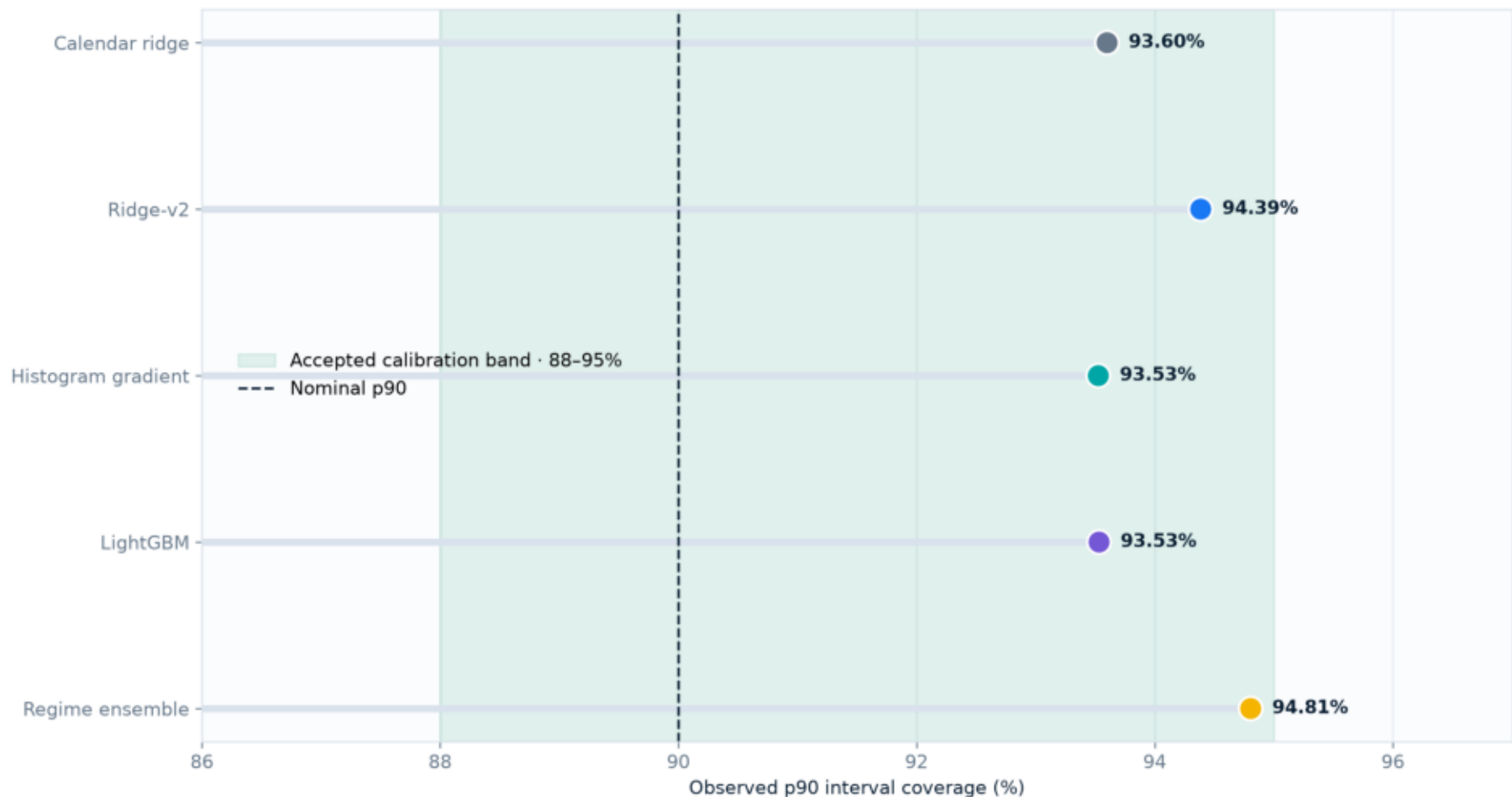
Causal event features materially reduced peak misses

Peak underprediction summed across scheduled-event and causally detected-surge episodes



Uncertainty calibration passed for every candidate

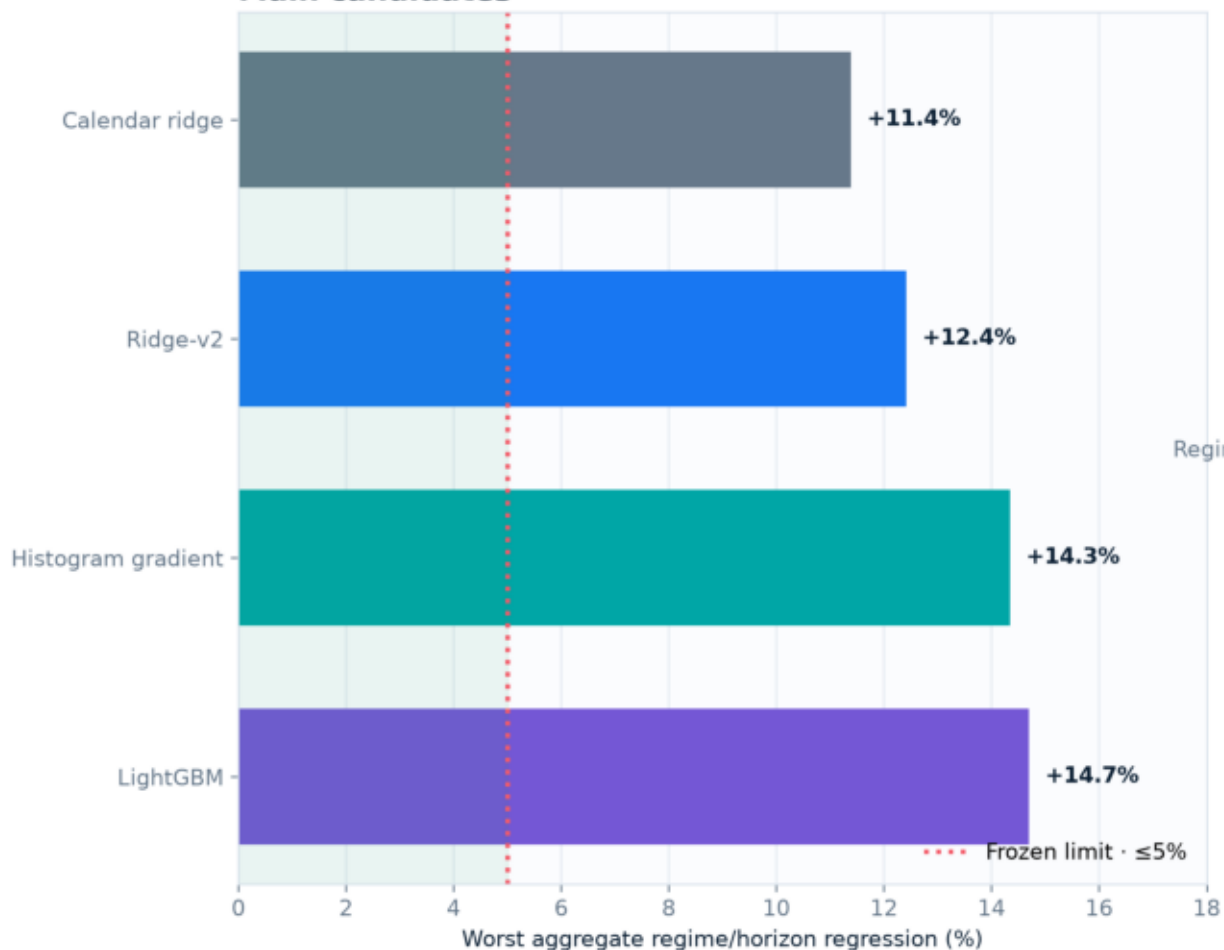
First-half residual calibration applied once; coverage measured only on the seed-46002 second half



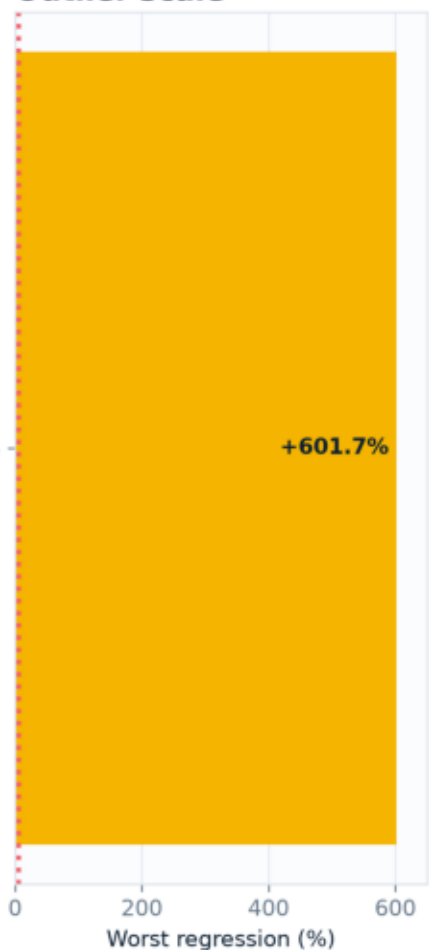
Tail regressions blocked promotion despite good average accuracy

Maximum weighted regression across every target x horizon x observable regime slice

Main candidates



Outlier scale



Synthetic seed-46002 selection · seed 46003 untouched

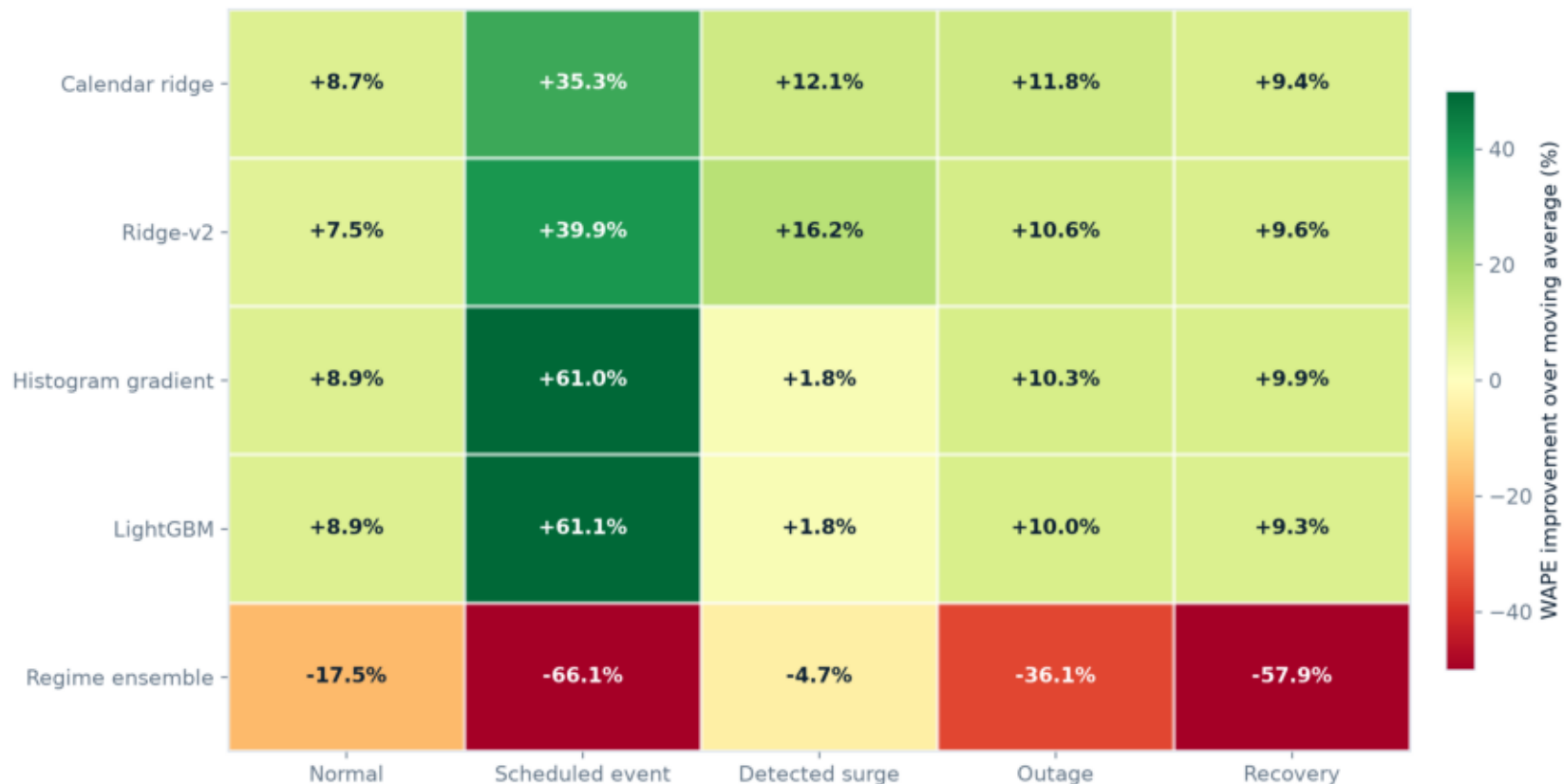
No model cleared the complete fail-closed selection contract

Promotion requires every column to pass; favourable averages cannot override a failed safety or causality gate

Calendar ridge	FAIL	FAIL	PASS	FAIL	PASS
Ridge-v2	FAIL	PASS	PASS	FAIL	PASS
Histogram gradient	FAIL	PASS	PASS	FAIL	PASS
LightGBM	FAIL	PASS	PASS	FAIL	PASS
Regime ensemble	FAIL	PASS	PASS	FAIL	PASS
	WAPE gain $\geq 15\%$	Peak-miss reduction $\geq 20\%$	p90 coverage 88-95%	Worst regression $\leq 5\%$	Causal surge scoring

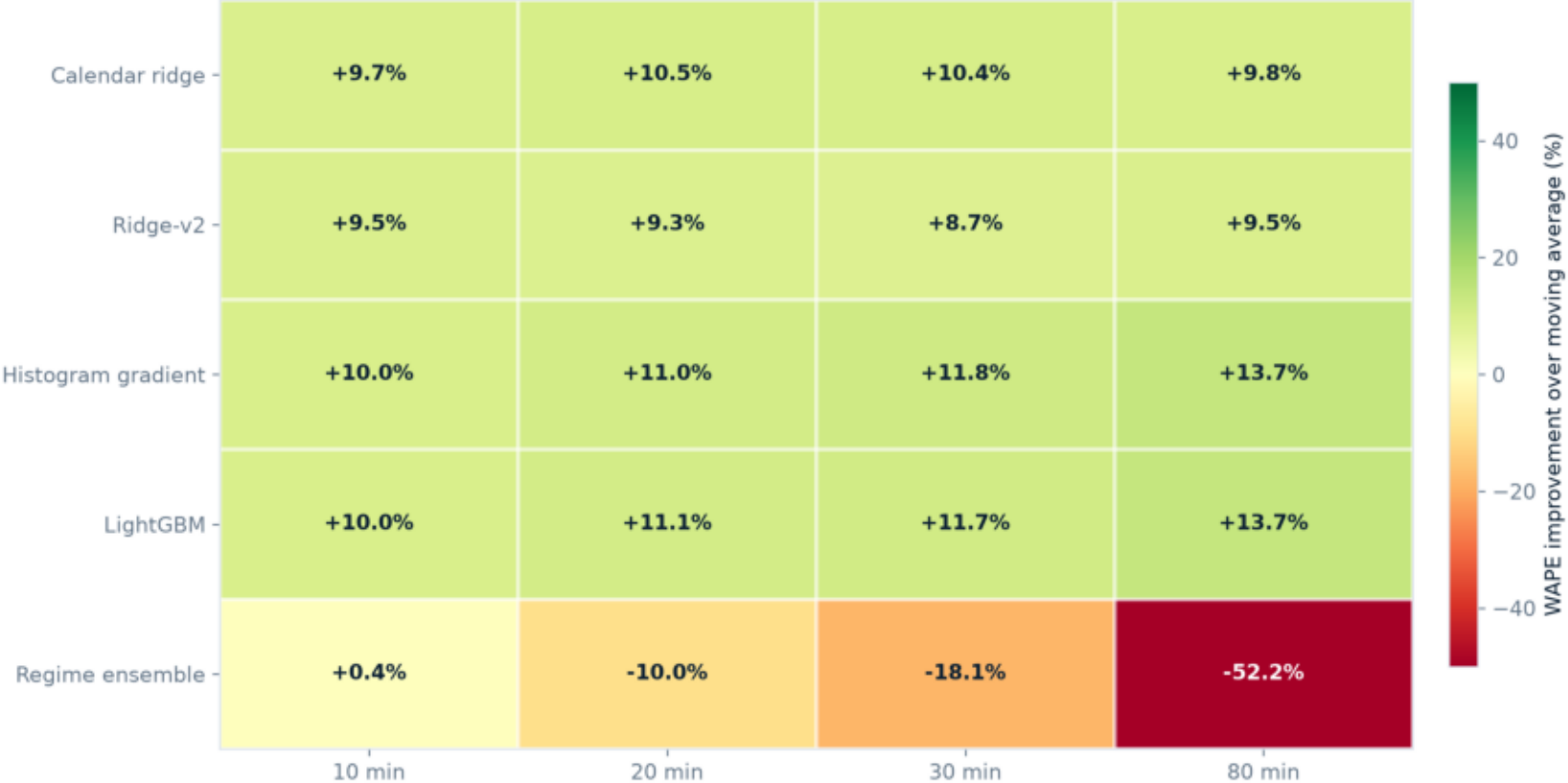
Average gains hide material regime dependence

Weighted WAPE improvement over the moving-average reference; values below -50% are colour-clipped, not numerically clipped



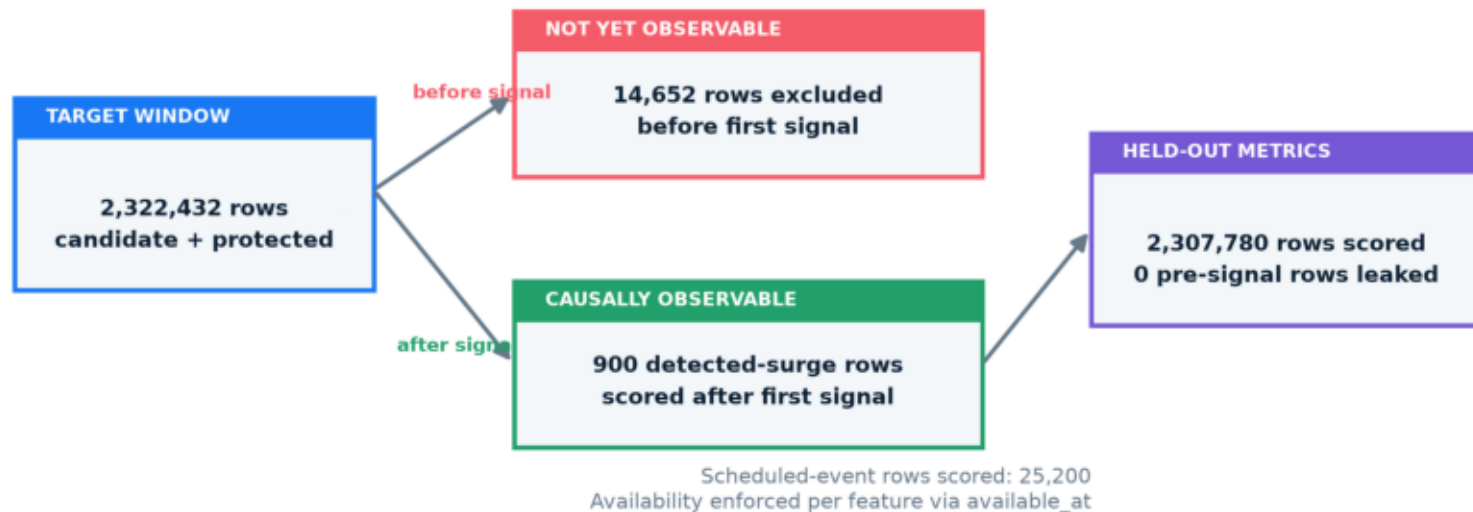
Forecast value changes across the planning horizon

Weighted across all groups, targets and observable regimes for each forecast horizon



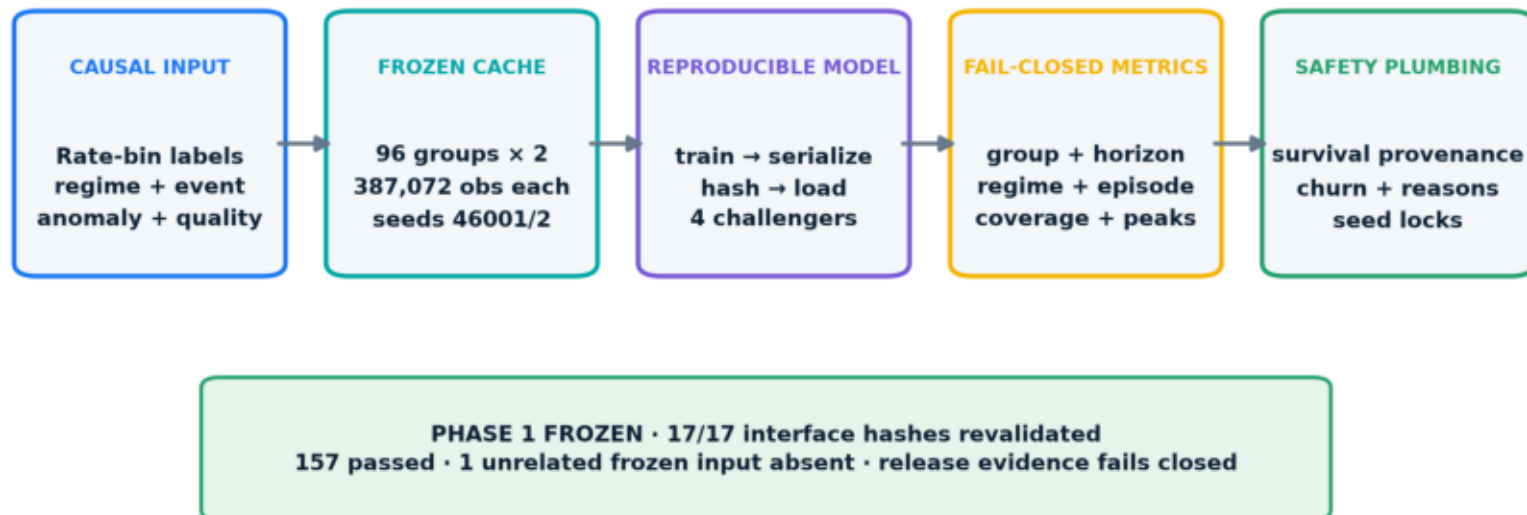
Unknown events become usable only after evidence exists

The offline selection replay applies the same causal availability contract as live observations



Phase 1 closed the experiment-plumbing gaps before model selection

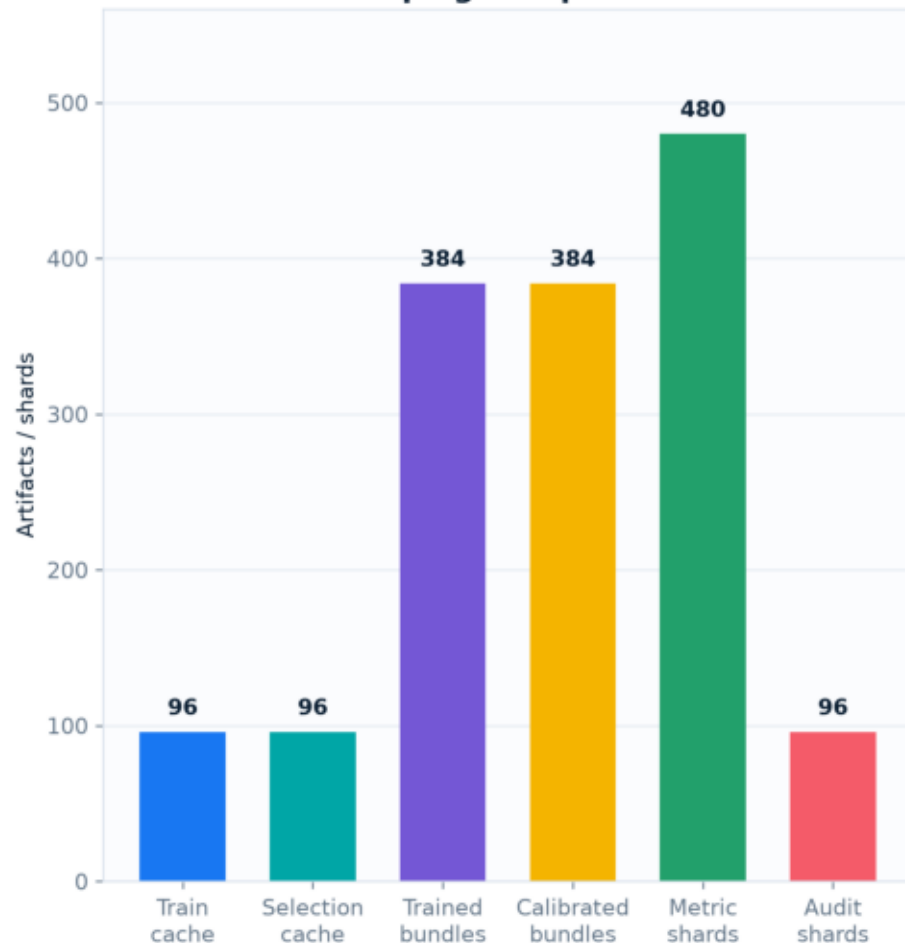
One observation contract now connects offline training, live forecasting, model packaging and controller evidence



The campaign was sharded for cluster-scale execution and independently audited

Counts come from the sealed Phase-2 completion record; cluster capacity is not presented as measured utilization

Authoritative campaign outputs



Cluster and release discipline

160 NODES

20,000 usable CPUs available

480 SELECTION SHARDS

Five 96-group arrays submitted concurrently

768 BUNDLES RELOADED

Every trained + calibrated challenger verified

0 PROTECTED TEST RUNS

Seed 46003 stayed behind the gate

The defensible deployment architecture is Static-first and event-gated

Forecast evidence informs a bounded controller role; it does not justify universal MPC or a challenger promotion

